

Modelling & Simulation: Assignment 3

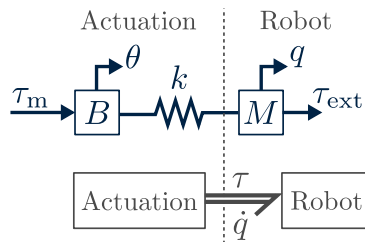
Series Elastic Actuators

Wesley Roozing, Geert Folkertsma

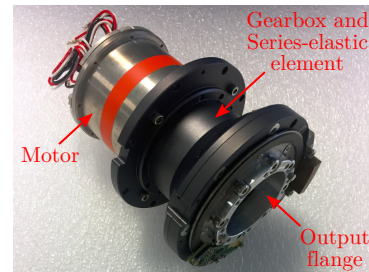
Introduction

In many situations in robotics, it is preferable not to do *position* control, but rather to apply desired *forces* on the robot or its environment. This is what enables robots to physically interact with their environment and humans, and to deal with uncertainty that arises outside of constrained factory environments. And physical interactions are a large part of what makes robots *useful*.

In this week's assignment you will work on modelling and simulation of a series-elastic actuator. Originating from *flexible joint robots*, that explicitly considered the finite drivetrain¹ stiffness of robot manipulators, the seminal work by Pratt et al. in 1995 proposed to add known flexibility to the drivetrain of robots *on purpose*. The resulting **series-elastic actuators** (SEAs, Fig. 1) allow for easier torque sensing and control², better robustness against physical impacts³, and higher energy efficiency in certain situations.



(a) A series-elastic actuator (SEA) with physical stiffness k driving a 1-DoF “robot” inertia M with configuration q . The motor rotor and gearbox inertia are given by an inertia B with configuration θ and its electromotive torque is a pure torque source τ_m . An external torque τ_{ext} (e.g. gravity) is acting on the load inertia.



(b) A real series-elastic actuator.

Figure 1: Series-elastic actuators.

The ability of SEAs to do accurate torque control makes them especially effective at implementing *impedance control*, a natural way of doing interaction tasks. Remember that *impedance* relates *effort* (force/torque) and *flow* (velocity), i.e. $H(s) = e(s)/f(s)$. For example, impedance control may make the actuator behave as a spring and damper as seen from the output.

¹*drivetrain*: the group of components that deliver mechanical power from the prime mover (i.e., the motor) to the driven components (i.e., the robot's limbs).

²Using Hooke's law, and encoders that measure the deflection of the elastic element, the torque control problem becomes a position control problem. Despite the low-pass filtering effect of the drivetrain compliance, torque control quality is usually much better, albeit limited in bandwidth.

³Gearboxes are easily damaged by high torque peaks, that result from inelastic collisions. Series elasticity effectively ‘filters’ such peaks mechanically.

Assignment

Starting from your previously modelled motor-gearbox combinations, model a series-elastic actuator with carefully chosen series stiffness and realistic friction, and implement both force and impedance control.

You do not need to account for the thermal domain, and you can assume your gearbox is infinitely stiff (i.e., a perfect speed reducer with friction).

Your report should include at least the following steps:

1. Start by modelling a series-elastic actuator based on the simple model shown in Fig. 1. Carefully consider how to design the stiffness value of the series-elastic element, that couples the gearbox to the output⁴. Assume that the load $M = 1 \text{ kg m}^2$.
2. Implement torque control, that allows you to track setpoints $\tau^*(t)$ for the torque τ that the actuator produces on the load inertia, even when the “robot” may be moving ($\dot{q} \neq 0$). Hint: Hooke’s law with a PD controller and maybe some feed-forward or compensation terms is a good start for decent performance.
3. Include a realistic friction model. Many friction models of varying levels of complexity could be implemented – Develop a model that includes (at least) Coulomb+viscous friction terms⁵. Show the influence of the friction effects on the actuator’s dynamics, and on the performance of your torque control.
4. Implement impedance control, that provides the torque reference τ^* for your torque controller (this means cascaded control loops!). Show that you can make the actuator behave as a spring-damper. Secondly, use impedance control to do positioning of the load.
5. Design a (repetitive) task for your actuators, and compare the efficiency of your two motor-gearbox combinations. Analyse where energy goes; how much is lost to electrical resistance and friction respectively, and how do your design parameters influence this? How would you change the design to improve/maximise efficiency?⁶

Suggested material:

- G. A. Pratt and M. M. Williamson, “Series elastic actuators”, IEEE/RSJ International Conference on Intelligent Robots and Systems, vol. 1, pp. 399–406, 1995.
- W. Roozing, J. Malzahn, N. Kashiri, D. G. Caldwell, and N. G. Tsagarakis, “On the Stiffness Selection for Torque-Controlled Series-Elastic Actuators”, IEEE Robotics and Automation Letters, vol. 2, no. 4, pp. 2255–2262, doi: 10.1109/LRA.2017.2726141, 2017.

⁴Temporarily fixing the actuator’s output in place ($\dot{q} \equiv 0$, or equivalently, $M \rightarrow \infty$) should help you to both design your SEA and develop control for it (2nd order system?).

⁵Coulomb friction depends on the *sign* of the velocity, that may lead to small oscillations around zero velocity. Is there a clever way to efficiently implement this in simulation?

⁶Your analysis and understanding is important here; not the best numbers.